FirstPass · Gemini benchmark

Letter of Credit document checking, tested on 20 presentations with real Gemini calls (no cache). AI Builder Cup 2026 · results of 14–15 Oct 2026

32 / 32	**0**	**0 of 6**	**12 / 12**
planted document discrepancies caught	false alarms (plain Gemini: 3)	clean cases wrongly flagged (plain Gemini: 2 of 6)	risky LC terms flagged before shipment

What was tested

Each case is a Letter of Credit (SWIFT MT700 layout) plus five shipping documents as PDFs: commercial invoice, bill of lading, packing list, certificate of origin and insurance certificate. 6 cases are clean; 14 have 1–3 planted discrepancies. The 20 cases were chosen so that together they contain **every planted rule type: 18 document checks and 8 LC-term checks**.

FirstPass: Gemini 3.8 Flash reads every page into a strict schema with an evidence box per field; a deterministic rules engine checks the documents against UCP 600 and the LC terms and against each other; Gemini 3.1 Pro judges wording. **Baseline:** one plain Gemini prompt per case with all page images and the list of discrepancy categories, no rules engine.

Results

	FirstPass	Plain Gemini prompt
Planted document discrepancies caught	**32 / 32 (100%)**	32 / 32 (100%)
False alarms (errors reported that were not there)	**0**	3
Clean presentations wrongly flagged	**0 of 6**	2 of 6 (33%)
Precision	**100%**	91.4%
Risky LC terms flagged before shipment	**12 / 12**, 0 false alarms	not offered
Evidence on the page, UCP 600 / MT700 reference, drafted fix	**every finding**	no

Plain Gemini finds errors well on clean digital PDFs. The difference is trust: it reported a wrong LC number on two compliant presentations and a missing document on a third. For an exporter, a false alarm means chasing a fix that is not needed; FirstPass reported none, and every finding it makes can be checked against the page.

Speed and cost

A fresh read of a six-document case took about 1–3 minutes (the LC about 17–40 s; the other documents are read in parallel). Re-checking the same documents comes from a cache keyed on the file and prompt: about a second, at no cost. Total Vertex AI spend on 13 Oct, covering setup, smoke tests and two 6-case runs, was about US$1.40. That suggests a few cents per presentation; check the exact figure on the billing page.

Limits of this test

The documents are generated with fictional parties and are clean digital PDFs, not phone photos or scans; real presentations will be harder. 4 of the 22 document rules have no planted case yet. UCP 600 article references should be checked against the official ICC text. * case_012, case_014 and case_021 are from a re-run on 15 Oct after two fixes: a cut-off Gemini reply on a packing list no longer fails the case, and the B/L consignee requirement is also read from the LC's field 46A. The other 17 cases do not touch the changed code.

Per case

Case	Planted discrepancies	FirstPass found	Plain Gemini found
case_000	clean	—	—
case_001	clean	—	**R24**
case_002	clean	—	—
case_003	clean	—	—
case_004	clean	—	**R24**
case_005	clean	—	—
case_010	R05 invoice over LC amount, R09 other doc description conflicts, R21 B/L consignee wrong	R05, R09, R21	R05, R09, R21
case_011	R05 invoice over LC amount	R05	R05
case_012*	R03 presentation period exceeded	R03	R03
case_013	R01 presented after LC expiry, R10 data conflict between documents	R01, R10	R01, R10
case_014*	R04 invoice currency differs, R07 invoice issuer / addressee wrong, R21 B/L consignee wrong	R04, R07, R21	R04, R07, R21
case_015	R05 invoice over LC amount, R11 port differs from LC, R20 country of origin differs	R05, R11, R20	R05, R11, R20
case_016	R11 port differs from LC	R11	R11
case_019	R03 presentation period exceeded, R04 invoice currency differs	R03, R04	R03, R04
case_020	R01 presented after LC expiry, R02 shipped after latest date, R14 insurance dated after shipment	R01, R02, R14	R01, R02, R14
case_021*	R04 invoice currency differs, R08 invoice description differs, R11 port differs from LC	R04, R08, R11	R04, R08, R11
case_022	R09 other doc description conflicts, R16 required document missing, R22 trade term differs	R09, R16, R22	R09, R16, R22
case_023	R10 data conflict between documents	R10	R10, **R16**
case_034	R10 data conflict between documents, R12 no on-board notation, R16 required document missing	R10, R12, R16	R10, R12, R16
case_035	R15 insurance amount / currency wrong, R21 B/L consignee wrong, R24 wrong LC number	R15, R21, R24	R15, R21, R24

Red = reported but not planted (false alarm).